

The Sovereign AI Operating System.

Custom kernel · on-device LLM · cryptographically signed actions.
Boots bare-metal — or coexists with Windows, macOS, Linux.

Custom Kernel

vos3.elf · deterministic build · KASLR

Bare Metal OR Hosted

boot direct · or sidecar Win/Mac/Linux

Local LLM

Llama · Mistral · Phi-3 GGUF Q4_K_M

Signed Actions

Ed25519 · X25519 · SQLCipher · HMAC

Problem · Solution

Why cloud-AI fails the highest-trust workloads — and what we built.

PROBLEM · CLOUD-AI LIABILITY

Every prompt is data exfiltration.

- SAP / EHR / IP / trades leak via cloud LLM inference.
- No cryptographic audit chain — "who approved that?"
- Air-gapped racks (defense · classified) locked out.
- EU AI Act + sovereignty mandates make it fineable.
- API rate limits & vendor lock-in.

SOLUTION · SOVEREIGN FABRIC

Custom kernel + signed actions.

- vos3.elf custom kernel · runs bare-metal or hosted.
- On-device LLM · 8 PUD-isolated AI inference slots.
- Ed25519-signed workflows · SQLCipher state · keyring.
- PermissionGate + Sandbox + Resource Guard.
- P2P mesh sync · no cloud required.

System Topology

The full process tree · how the 4 components communicate.

FRONTEND

Next.js 16 · React 19 · Tailwind v4 · @xyflow · framer-motion

Workflow Builder · Approval Drawer · P2P Node Map · Sovereign Audit Trail

frontend/

HTTPS / WebSocket · Clerk · Bearer auth · CSRF token

BACKEND

FastAPI · Python 3.12 · SQLAlchemy 2.0 · Pydantic

PermissionGate · SmartRouter · Orchestrator · HostBridge · P2P · Crypto

backend/

VBus · Unix socket / virtio · HMAC-SHA256 frame · CRC32C

KERNEL

vos3.elf · C · x86_64 · limine bootloader

KIM AI · Crypto Engine · Memory MGR · Scheduler · VFS · AI Guard

kernel/build/

spawn / mmap · QEMU child · ivshmem 4x16 MB warp zones

SHELL

Tauri 2.10 · Rust · QEMU child process

WebView host · ivshmem mmap · biometric prompt · installer (.dmg/.msi/.ApplImage)

desktop/src-tauri/

Kernel Architecture · vos3.elf

What lives inside the kernel ELF · subsystems · silicon hardening.



Security Model · Permissions

5 enforcement layers · every breach fails closed · every action is audited.

Manifest Scopes

grammar: host.automation · network.egress:host · llm.local

L1

- Hierarchical: granted="llm" implies requested="llm.cloud"
- Restrictions beat grants (asymmetric — denial wins)
- manifest_json column · signed by developer (Ed25519)

PermissionGate

backend/services/app_sandbox.py:495-542 · cached in-memory

L2

- Every gate.check(app_id, *scopes) call inspects manifest
- Cache hydrated from SQLite · invalidated on UPDATE
- Denial → _record_security_event(scope_violation) → 403

Sandbox + Resource Guard

AppProcessRunner · ulimit envelope · ms-fast classifier

L3

- RLIMIT_AS=128MB · RLIMIT_CPU=5s · RLIMIT_NOFILE=64
- SIGXCPU/SIGKILL/SIGSEGV → _classify_resource_violation
- Auto-isolate BEFORE return → no concurrent race window

Silicon-Backed Crypto

crypto_keyring.py · macOS Keychain · DPAPI · SecretService

L4

- Fernet HKDF-SHA256 fallback bound to machine-id
- SQLCipher PRAGMA key from keyring · 256-bit
- Workflow runs Ed25519-signed · tamper auto-quarantines

HITL Transaction Guard

host_bridge.py · TransactionGuard · pendingHostApprovals

L5

- Default cap: \$5K · rate: 10/min · per-(app,target) window
- Hold → operator signs canonical bytes with Ed25519
- Public key pinned to keyring — fresh keypair rejected

SINGLE SOURCE OF TRUTH · securityAuditLog table

audit kinds written by every layer:

scope_violation · path_traversal_attempt · app_isolated · manifest_tampered · workflow_tampered · resource_limit_exceeded · local_fallback_active · malicious_peer_a

Local AI Pipeline

Kernel-side inference · Smart Router · Ollama bridge · supported models.

KIM · KERNEL AI INFERENCE

kernel/src/mm/ai_kim.c · kernel/src/ai/ggml_core.c

- 8 slots (Slot 0 = Coordinator) — PUD-isolated.
- Per-slot KV cache: 4 × 2 MB HugePages = 8 MB.
- Working buffers: hidden ≤ 32 KB · logits ≤ 256 KB.
- GGUF v2/v3 parser · 4096 max tensors.
- Weights mapped RO into slot via Warp ivshmem.
- VBus KIM_GENERATE|slot|max_tokens|temp_fp100
- Token stream: VBUS_TYPE_TOKEN_STREAM 0x07

SMART ROUTER · MODEL CATALOG

backend/config/router.yaml · backend/src/efficiency.py

Complexity-gated · threshold = 9 · local-first preference

1. kernel-default (in-kernel KIM slot)
2. local-default (ollama daemon)
3. claude-opus (cloud · complexity ≥ 9)
4. gpt-4o / gpt-codex (cloud · architect)
5. gemini-flash/pro (cloud · researcher)

Cost reduction vs pure cloud: 40-60% (telemetry-tracked).

SUPPORTED MODELS — kernel ceiling: vocab ≤ 65 536 · d_model ≤ 8192 · seq ≤ 4096

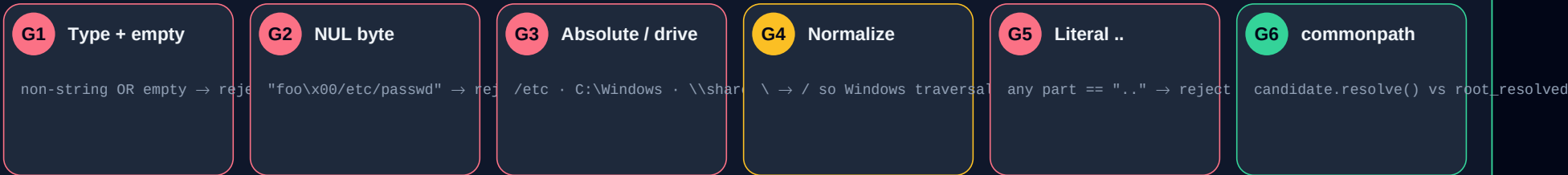
Model	Layers	d_model	Vocab	Q4 size	Kernel?
Llama-2-7B-chat Q4_K_M	32	4096	32 000	≈4 GB	✓
Mistral-7B-Instruct v0.2 Q4_K_M	32	4096	32 000	≈4 GB	✓
Llama-2-13B-chat Q4_K_M	40	5120	32 000	≈7.5 GB	✓
Phi-3-mini-4k Q4_K_M	32	3072	32 064	≈2.4 GB	✓
Llama-3-8B / 70B	32+	4096+	128k	—	✗ (vocab)
Gemma-2-9B / Qwen-2.5-7B	42	3584	152k+	—	✗ (vocab)

Ollama bridge handles models exceeding kernel ceilings (Llama-3, Gemma, Qwen) via the host daemon.

Filesystem Sandbox · Resource Guard

`_resolve_inside_sandbox` · `ulimit envelope` · `OOM-Kill lifecycle`.

`_resolve_inside_sandbox` — 6 gates · `backend/services/app_filesystem.py:152-228`



AUTO-ISOLATION CHAIN (any gate fail)

1. `_record_security_event(kind="path_traversal_attempt", details={...,"auto_isolated":true})`
2. `SANDBOX_MANAGER.isolate(app_id)` → UPDATE apps SET status='isolated'
3. `PERMISSION_GATE._clear(app_id)` → cache eviction (no stale allow)
4. `raise PathTraversalAttempt (PermissionError)` → route layer translates to HTTP 403

TOCTOU defense: final open uses `os.open(O_RDONLY|O_NOFOLLOW)` — kernel rejects symlink races.

RESOURCE ENVELOPE — `AppProcessRunner`

<code>RLIMIT_AS</code>	128 MB	(<code>DEFAULT_MEM_MB</code>)
<code>RLIMIT_CPU</code>	5 s soft / 10 s hard	
<code>RLIMIT_NPROC</code>	4	(basic thread/pipe)
<code>RLIMIT_NOFILE</code>	64	
<code>RLIMIT_FSIZE</code>	10 MB	
Concurrency	<code>asyncio.Semaphore(4)</code>	
Env allow	<code>PATH · HOME · LANG · VOS3_APP_*</code>	

OOM-KILL LIFECYCLE — ms-by-ms

<code>t=0</code>	child <code>malloc()</code> exceeds <code>RLIMIT_AS</code> → <code>ENOMEM</code>
<code>t≈1ms</code>	Python raises <code>MemoryError</code> (or kernel <code>SIGKILL/SIGXCPU</code>)
<code>t≈5ms</code>	runner returns <code>ExecutionResult</code> to caller
<code>t≈6ms</code>	<code>_classify_resource_violation</code> inspects <code>rc + stderr</code>
<code>t≈7ms</code>	<code>SANDBOX_MANAGER.isolate(app_id)</code> — gate cache cleared
<code>t≈8ms</code>	<code>_record_security_event(kind="resource_limit_exceeded")</code>
<code>t≈10ms</code>	next <code>gate.check()</code> sees status='isolated' → 403

P2P Mesh · Cryptographic Stack

UDP-signed beacons · X25519 ECDH · HKDF→Fernet · Ed25519 workflows.

DISCOVERY · LocalPeerDiscovery

backend/services/p2p_discovery.py

Beacon envelope · UDP 13370 · ≤ 1024 B

```
{
  "v": 1,
  "node_id": "<uuid4>",
  "workspace_id": "<workspace>",
  "sync_port": 13371,
  "ts_ms": 1715600000000,
  "public_key": "<64 hex · Ed25519>",
  "signature": "<128 hex · canonical>"
}
```

Verifier path · rejection branches

- signature invalid → malicious_peer_attempt
- workspace mismatch → local_tampering_blocked
- self echo → drop silently

HANDSHAKE · SovereignSyncEngine

backend/services/p2p_sync.py · TCP framed

4-step protocol

1. client.hello → ephemeral_pk(X25519) +
identity_pk(Ed25519) +
Ed25519 sig(ephemeral_pk■ws_id)
2. server.hello → same shape
3. mutual verify (both signatures + workspace)
4. ECDH: shared = X25519(priv, peer_pub)
transcript: sorted(a,b) → "a|b"
key = HKDF-SHA256(shared, info=transcript)
urlsafe-b64 → Fernet symmetric key

Failure paths (no encrypted bytes ever exchanged):

- signature invalid → malicious_peer_attempt audit
- workspace mismatch → local_tampering_blocked audit
- raises HandshakeFailed → socket torn down immediately

WORKFLOW SIGNATURE CHAIN · Ed25519 over canonical bytes

backend/services/agent_orchestrator.py · signing_public_key column on workflowRuns

```
submit_workflow() →
  priv, pub = get_or_mint_workflow_signing_key() # P6.1 keyring-pinned
  canon = canonical_bytes({run_id, workspace_id, manifest_json, created_at_ms})
  sig = Ed25519(priv).sign(canon)
  INSERT INTO workflowRuns (... , signature, signing_public_key) VALUES (...)
```

on every _load / run_status:

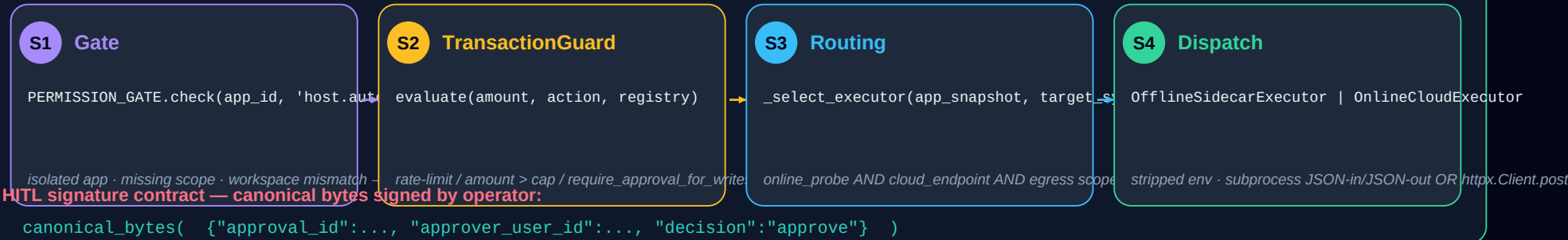
```
if not verify_manifest_signature(canon, row.signature, row.signing_public_key):
  row.status = 'failed' · row.error = 'signature mismatch'
  _record_security_event(kind='workflow_tampered', ...)
  raise WorkflowTampered(run_id) → caller catches → run quarantined
```

Host Bridge · SAP Automation Example

HostBridgeService · TransactionGuard · HITL approval flow.

HostBridgeService.execute_host_automation — 4 sequential stages

backend/services/host_bridge.py:480-720



SAP GUI AUTOMATION — end-to-end signed flow



Deployment Modes · Hardware · Stack Summary

Where it runs · what it runs on · what it's built from.

MODE A · HOSTED

vos3.elf as Tauri sidecar

- QEMU child process · ivshmem warp zones.
- Coexists with Win/Mac/Linux + their apps.
- Default for offices · operator workstations.
- Network: kill-switch + manifest-gated egress.

MODE B · BARE METAL

vos3.elf IS the operating system

- limine 4.x bootloader · multiboot2 · KASLR.
- No host OS → zero host attack surface.
- Defense · classified racks · sovereign DC.
- Same kernel binary — only the boot path differs.

HARDWARE TIERS

MIN

CPU: x86_64 / Apple M1
RAM: 8 GB
Storage: 5 GB
OS: macOS 12 · Win 10 · Linux 5.10
Model: Phi-3-mini Q4 (≈3 GB)

REC

CPU: AVX2 / Apple M2 · M3
RAM: 16-32 GB
Storage: 50 GB SSD
OS: macOS 14 · Win 11 · Ubuntu 22
Model: Llama-7B / Mistral-7B Q4_K_M

ENT

CPU: AVX-512 / M3 Pro · NPU
RAM: 64+ GB
Storage: 500 GB NVMe
+ TPM 2.0 / Apple Secure Enclave
Model: Llama-13B Q4 · P2P mesh

METAL

CPU: x86_64 · multiboot2 capable
RAM: 16 GB (32 GB ideal)
Storage: 10 GB · UEFI partition
OS: none (vos3.elf primary)
Hardware: TPM 2.0 attestation

STACK SUMMARY

Layer	Stack	Path
Kernel	C · x86_64 · limine 4.x · 30k SHA-256 freestanding	kernel/build/vos3.elf
Backend	Python 3.12 · FastAPI · SQLAlchemy 2.0 · Pydantic v2	backend/
Database	SQLite + SQLCipher (Fortress) · Convex (Online)	core/database/
Crypto	cryptography (Ed25519 · X25519 · HKDF · Fernet)	services/app_crypto.py · crypto_keyring.py
Frontend	Next.js 16 · React 19 · Tailwind v4 · @xyflow/react	frontend/
Shell	Rust 2024 · Tauri 2.10 · QEMU child sidecar	desktop/src-tauri/
AI · local	GGUF Q4_K_M · KIM kernel slots · Ollama bridge	kernel/src/ai/ · backend/ai/
Compliance	EU AI Act mapped · SOC 2 roadmap · ISO 42001	docs/EU_AI_ACT_COMPLIANCE.md